Beyond Task Success: Measuring Workflow Fidelity in LLM-Based Agentic Payment Systems

Donghao HUANG[1,2][0009-0005-6767-4872], Joon Kiat CHUA[1][0009-0001-8333-9792], and Zhaoxia WANG[1,⋆][0000-0001-7674-5488]

[1] School of Computing and Information Systems, Singapore Management University, 80 Stamford Rd, Singapore 178902, Singapore
[2] Research and Development, Mastercard, 4250 Fairfax Dr, Arlington, VA 22203, USA

Abstract. LLM-based multi-agent systems are increasingly deployed for payment workflows, yet prevailing metrics—Task Success Rate (TSR) and Agent Handoff F1-Score (HF1)—capture only final outcomes or unordered routing decisions. We introduce the *Agentic Success Rate* (ASR), a trajectory-fidelity metric that compares observed and expected agent execution sequences at the *transition* (bigram) level, decomposing performance into Transition Recall (completeness) and Transition Precision (efficiency). Applied to the Hierarchical Multi-Agent System for Payments (HMASP) across 18 LLMs and 90,000 task instances (5 repeats per model–scenario pair), ASR reveals that 11 of 18 models systematically skip a confirmation checkpoint during payment checkout—invisible to both TSR and HF1—while 7 models enforce the checkpoint perfectly. Notably, 12 open-weight models outperform both proprietary GPT models on average ASR. Prompt refinements and deterministic routing guards guided by ASR diagnostics yield substantial TSR improvements, with gains up to +93.8 pp for previously struggling models, demonstrating that trajectory-level evaluation is essential in regulated domains.

Keywords: Agentic Success Rate · Agentic Payments · LLM Multi-Agent Systems · Evaluation Metrics · Digital Finance

1 Introduction

The emergence of autonomous LLM-based agents has fueled rapid adoption of *agentic payments*—using LLM agents to automate end-to-end payment workflows [16]. Major payment networks have launched agentic capabilities: Mastercard's Agent Pay [12] and Visa's Intelligent Commerce [17], with projections reaching $1.7 trillion by 2030 [16]. In research, frameworks like HMASP [5] have demonstrated the feasibility of hierarchical multi-agent systems for payments.

Despite these deployments, current evaluation frameworks remain inadequate. HMASP, for example, relies on Task Success Rate (TSR) and Agent

⋆ Corresponding author: zxwang@smu.edu.sg

Handoff F1-Score (HF1). However, TSR captures only final outcomes, and HF1 evaluates routing in a bag-of-edges fashion, ignoring temporal order. Neither captures *workflow fidelity*—the agent's adherence to the expected sequence of steps. In payment systems, auditability is critical [14], and the sequential path to the result is as important as the outcome itself [4, 3].

Drawing on process mining conformance checking [1, 2], we propose the **Agentic Success Rate (ASR)**, a trajectory-level metric that compares agent-to-agent *transitions* (consecutive handoff pairs) between observed and expected workflows. Our contributions are: (1) ASR as a multi-agent payment metric; (2) evaluation of 18 LLMs on 90,000 instances; (3) discovery of workflow shortcuts invisible to TSR/HF1; and (4) demonstration that ASR-guided engineering yields dramatic TSR improvements for previously struggling models.

2 Related Work

LLM-Based MAS for Payments. Multi-agent systems have been applied to software engineering [8], trading [18], and investment [20]. In payments, prior work addressed process automation [7] and fraud detection [6] but not end-to-end agentic workflows. HMASP [5] filled this gap with a hierarchical agent framework.

Evaluating Agentic Systems. Standard evaluation focuses on task completion [10, 9], but binary metrics obscure behavioral nuances. AgentBoard [11] introduced Progress Rate; WorFEval [15] uses subsequence matching for workflow planning; τ-bench [19] proposed pass^k for reliability; and Chen et al. [4] argued standard benchmarks underestimate financial risks. ASR builds on this line of work [13], grounding trajectory evaluation in process mining conformance checking [1, 2].

3 Methodology

3.1 Agentic Success Rate (ASR)

ASR compares the full ordered trajectory against the expected one at the *transition* level, drawing on the conformance checking paradigm from process mining [1, 2]. Each consecutive pair of agents in a trajectory is a transition (bigram); for example, the trajectory $[A, B, C]$ yields the transition multiset $\{(A,B),\ (B,C)\}$. Given expected trajectory $E = [e_1, \ldots, e_n]$ and observed trajectory $O = [o_1, \ldots, o_m]$, let B_E and B_O denote their respective transition multisets:

$$\mathrm{TR} = \frac{|B_E \cap B_O|}{|B_E|}, \quad \mathrm{TP} = \frac{|B_E \cap B_O|}{|B_O|}, \quad \mathrm{ASR} = F_1(\mathrm{TR}, \mathrm{TP}) \tag{1}$$

An ASR of 100% indicates perfect transition alignment. Low Transition Recall (TR) signals skipped handoffs ("model moves" in process mining terminology); low Transition Precision (TP) signals redundant ones ("log moves"). Because

ASR operates on transitions rather than individual agent visits or unordered edge sets, it directly measures whether each agent handed off control to the correct next agent—precisely the property needed for compliance auditing—and generalizes to arbitrary agentic architectures.

3.2 Evaluation Setup

Models. We evaluated 18 models spanning both open-weight and proprietary LLMs. From the original HMASP study [5], we include Qwen3 (8B, 14B, 32B), Qwen2.5 (7B, 14B, 32B), Mistral-Small-3.2 (24B), Magistral (24B), Llama 3.1 (8B, 70B), and GPT-4.1. Additionally, we evaluated seven models not included in the original study: GPT-5.2, GPT-OSS (20B, 120B), and Gemma4 (E2B, E4B, 26B, 31B). Proprietary models (GPT-4.1, GPT-5.2) are accessed via the official OpenAI API, while all open-weight models are run locally using Ollama v0.20.0 on an Apple M5 Max with 128 GB unified memory and 4-bit quantization (MXFP4 for GPT-OSS models; Q4_K_M for all others).
Dataset and Repeat Protocol. We use the HMASP evaluation dataset of 1,000 data points across four scenarios (250 each): **T1** (card registration; e.g., *"Add a new card"*), **T2** (card retrieval; e.g., *"Show my saved cards"*), **T3** (payment processing; e.g., *"Complete my purchase"*), and **T4** (irrelevant input rejection; e.g., *"What's the weather today?"*). Each model–scenario pair is evaluated 5 times independently, yielding 90,000 total instances. We report the mean and standard deviation across the 5 repeats.
Metrics. We report three metrics, the first two from the original HMASP evaluation [5]: **(1) Task Success Rate (TSR):** the proportion of runs completing the correct workflow with correct information saved or retrieved; **(2) Agent Handoff F1-Score (HF1):** the F1-score comparing actual vs. expected agent-to-agent handoffs; and **(3) Agentic Success Rate (ASR):** trajectory-level workflow fidelity as defined above.
System Engineering. We improve the HMASP system through two complementary techniques, both guided by ASR diagnostics. **(i) Prompt engineering:** agent prompts are refined iteratively to better specify roles, expected handoffs, and task boundaries. **(ii) Deterministic routing guards:** a *cart-items guard* in the payment supervisor suppresses the payment-review path when `cart_items` is empty, and *card-view pattern matching* in the CPA provides keyword-based fallback routing for ambiguous card intents.

4 Results and Discussion

4.1 Effectiveness of System Engineering

Table 1 compares TSR between HMASP [5] and our system for the three most-improved models (ranked by average Δ across all tasks). The gains are striking: Llama3.1:8b improves by an average of +67.9 pp, Magistral:24b by +54.2 pp, and Llama3.1:70b by +33.5 pp—confirming that failures were addressable through engineering rather than model replacement.

Table 1. HMASP vs. Ours: TSR (%) for the three most-improved models. H = HMASP, O = Ours (mean over 5 repeats), Δ = O - H. Bold: $\geq$20 pp gains.

	T1			T2			T3			T4		
Model	**H**	**O**	**Δ**	**H**	**O**	**Δ**	**H**	**O**	**Δ**	**H**	**O**	**Δ**
Llama3.1:8b	6.0	99.8	+**93.8**	10.4	98.8	+**88.4**	4.0	93.4	+**89.4**	100	100	+0.0
Magistral:24b	44.0	90.0	+**46.0**	18.4	97.7	+**79.3**	3.2	94.7	+**91.5**	100	100	+0.0
Llama3.1:70b	66.0	100	+**34.0**	74.0	100	+**26.0**	26.0	98.9	+**72.9**	98.8	100	+1.2

4.2 ASR Reveals Hidden Workflow Deviations

Table 2 reports T3 metrics (mean $\pm$ std over 5 repeats) and average across T1–T4 for all 18 models, sorted by average ASR. Seven models achieve a perfect 100% on all three T3 metrics: all four Gemma4 variants, both GPT-OSS models, and `MSmall3.2:24b`. A striking result is that **12 open-weight models outperform the proprietary GPT-4.1 and GPT-5.2** on average ASR. Among the 11 models that exhibit a T3 ASR gap, the range spans from 0.01% (`Qwen2.5:32b`) to 5.45% (`Qwen2.5:7b`); notably, `GPT-4.1` and `GPT-5.2` show the largest gaps among high-performing models at 2.90% and 2.72% respectively. The highlighted cells are the sharpest examples: `Qwen2.5:32b`, `Qwen3:32b`, and `Qwen3:8b` achieve T3 TSR = HF1 = 100% yet ASR < 100%, exposing skipped confirmation checkpoints invisible to conventional metrics.

Table 2. T3 (Payment) results (%, mean $\pm$ std over 5 repeats) and average across T1–T4 for all 18 models. Sorted by Avg ASR desc. Yellow : TSR = HF1 = 100% but ASR < 100%.

	T3 (Payment)			Avg (T1–T4)		
Model	**TSR**	**HF1**	**ASR**	**TSR**	**HF1**	**ASR**
6 models[†]	100	100	100	100.0	100.0	100.0
Qwen2.5:32b	100	100	99.99±0.02	100.0	100.0	100.0
Qwen3:32b	100	100	99.97±0.04	100.0	100.0	100.0
Qwen3:14b	99.7±0.3	99.8±0.2	99.6±0.3	99.9	100.0	99.9
Qwen3:8b	100	100	99.97±0.02	99.9	100.0	99.9
Llama3.1:70b	98.9±1.2	98.8±1.1	98.8±1.2	99.7	99.7	99.7
Qwen2.5:14b	99.8±0.2	99.8±0.2	99.6±0.2	99.3	99.6	99.6
GPT-4.1	99.8±0.4	99.9±0.2	97.1±0.3	99.5	99.7	98.8
GPT-5.2	99.9±0.2	100.0±0.1	97.3±0.1	99.4	99.7	98.8
Gemma4:e2b	100	100	100	98.4	98.8	98.4
Llama3.1:8b	93.4±0.9	96.1±0.7	93.3±1.0	98.0	98.8	98.0
Magistral:24b	94.7±1.2	97.1±0.6	93.9±1.0	95.6	97.7	95.5
Qwen2.5:7b	53.3±2.1	64.2±2.3	47.8±2.0	77.5	85.1	77.1

[†]Gemma4:e4b/26b/31b, GPT-OSS:20b/120b, MSmall3.2:24b.

The root cause is consistent across all 11 models exhibiting a T3 ASR gap. The expected T3 trajectory has 11 agent hops (10 transitions), including an intermediate return to the CPA for explicit user *confirmation* before payment processing. For direct payment phrases (e.g., "Process my payment"), models infer unambiguous intent and skip this checkpoint, executing 9 hops (8 transitions). Of 10 expected transitions, 8 appear: TR = 8/10 = 80%, TP = 8/8 = 100%,

ASR $= F_1(80\%, 100\%) = 88.9\%$ per affected sample. Crucially, every single hidden deviation across all affected models follows this *identical* 9-step shortcut—there is exactly one deviant trajectory pattern, reflecting a systematic interaction between input phrasing and model reasoning rather than stochastic variation. This has regulatory significance: PCI-DSS [14] requires auditable workflows, and bypassing confirmation checkpoints creates gaps in the audit trail even when outcomes are correct.

5 Conclusion

We introduced ASR as a trajectory-level evaluation metric for LLM-based multi-agent payment systems. Across 18 LLMs and 90,000 instances, ASR exposes a systematic procedural deviation: 11 of 18 models skip a CPA-mediated confirmation checkpoint during payment checkout (T3 ASR gaps of 0.01–6.7%) despite near-perfect TSR and HF1, while 7 models enforce the checkpoint perfectly. Notably, 12 open-weight models outperform both proprietary GPT models on average ASR, with `GPT-4.1` and `GPT-5.2` showing the largest gaps among high-performing models. ASR-guided system engineering yields dramatic improvements for previously struggling models (up to +93.8 pp). As payment networks deploy agentic capabilities at scale [12, 17], ASR provides the visibility needed to ensure procedural fidelity—essential where compliance, auditability, and liability attribution are at stake. Future work includes validating ASR in production systems, weighted variants for security-critical transitions, architectural enforcement of the T3 checkpoint, and cross-architecture evaluations [17].

References

1. van der Aalst, W.M.P., Adriansyah, A., van Dongen, B.F.: Replaying history on process models for conformance checking and performance analysis. WIREs Data Mining and Knowledge Discovery **2**(2), 182–192 (2012)
2. Carmona, J., van Dongen, B., Solti, A., Weidlich, M.: Conformance Checking: Relating Processes and Models. Springer (2018)
3. Cemri, M., Pan, M.Z., Yang, S., Agrawal, L.A., Chopra, B., Tiwari, R., Keutzer, K., Parameswaran, A., Klein, D., Ramchandran, K., et al.: Why do multi-agent LLM systems fail? (2025)
4. Chen, Z., Bhatia, A., Zhang, S.L., Choi, S., Saar-Tsechansky, M., Ghassemi, M.: Standard benchmarks fail – auditing LLM agents in finance must prioritize risk. arXiv preprint arXiv:2502.15865 (2025)
5. Chua, J.K., Huang, D., Wang, Z.: A novel hierarchical multi-agent system for payments using LLMs. Proceedings of the 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD) (2026), main Conference
6. Dahiphale, D., Madiraju, N., Lin, J., Karve, R., Agrawal, M., Modwal, A., Balakrishnan, R., Shah, S., Kaushal, G., Mandawat, P., et al.: Enhancing trust and safety in digital payments: An LLM-powered approach. In: 2024 IEEE International Conference on Big Data (BigData). pp. 4854–4863. IEEE (2024)

7. Guan, Y., Wang, D., Chu, Z., Wang, S., Ni, F., Song, R., Zhuang, C.: Intelligent agents with LLM-based process automation. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. pp. 5018–5027. KDD '24, Association for Computing Machinery (2024). https://doi.org/10.1145/3637528.3671646
8. Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Wang, J., Zhang, C., Wang, Z., Yau, S.K.S., Lin, Z., Zhou, L., Ran, C., Xiao, L., Wu, C., Schmidhuber, J.: MetaGPT: Meta programming for a multi-agent collaborative framework. In: The Twelfth International Conference on Learning Representations (2024), https://openreview.net/forum?id=VtmBAGCN7o
9. Kapoor, S., Narayanan, A., et al.: AI agents that matter. arXiv preprint arXiv:2407.01502 (2024)
10. Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., et al.: AgentBench: Evaluating LLMs as agents. arXiv preprint arXiv:2308.03688 (2023)
11. Ma, C., Zhang, J., Long, Z., Xie, Z., Chen, J., Zheng, J., et al.: AgentBoard: An analytical evaluation board of multi-turn LLM agents. In: Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track (2024)
12. Mastercard: Mastercard agent pay – powering the next frontier of commerce. https://www.mastercard.com/global/en/business/artificial-intelligence/mastercard-agent-pay.html (2025), accessed: 2025-11-15
13. Mohammadi, M., Rahmani, H.A., Nguyen, T.T., Ranasinghe, T., Macdonald, C., Ounis, I.: Evaluation and benchmarking of LLM agents: A survey. In: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (2025). https://doi.org/10.1145/3711896.3736570
14. PCI Security Standards Council: Ai principles: Securing the use of ai in payment environments. https://blog.pcisecuritystandards.org/ai-principles-securing-the-use-of-ai-in-payment-environments (2025), accessed: 2025-11-15
15. Qiao, S., Fang, R., Qiu, Z., Liu, X., Cheng, J., Chen, H., Zhang, N.: Benchmarking agentic workflow generation. In: The Thirteenth International Conference on Learning Representations (ICLR) (2025), https://openreview.net/forum?id=vunPXOFmoi
16. Schumacher, K., Roberts, R., Giebel, K.: The agentic commerce opportunity: How AI agents are ushering in a new era for consumers and merchants. https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-agentic-commerce-opportunity-how-ai-agents-are-ushering-in-a-new-era-for-consumers-and-merchants (2025), accessed: 2025-11-15
17. Visa: Enabling AI agents to buy securely and seamlessly. https://corporate.visa.com/en/products/intelligent-commerce.html (2025), accessed: 2025-11-15
18. Xiao, Y., Zhao, M.K., Zhou, R., Boen, J.: TradingAgents: Multi-agents LLM financial trading framework. arXiv preprint arXiv:2412.20138 (2024)
19. Yao, S., Narang, R., Bhatt, A., Chen, W., Bisk, Y.: τ-bench: A benchmark for tool-agent-user interaction in real-world domains. arXiv preprint arXiv:2406.12045 (2024)
20. Yu, Y., Yao, Z., Li, H., Deng, Z., Jiang, Y., Cao, Y., Chen, Z., Suchow, J., Cui, Z., Liu, R., et al.: Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making. Advances in Neural Information Processing Systems **37**, 137010–137045 (2024)